作业 4

1 . 贮水系，数贮水数，给水度，渗透系数，导水系数，压力传导系数，越流系数，越流因素

2 .

(1) 若取 K_1 方向为坐标轴 x ，K_2 方向为坐标轴 y，此时渗透系数张量为主对角阵

$$K = \begin{pmatrix} K_1 & 0 \\ 0 & K_2 \end{pmatrix}$$

设含水层厚度 M, 则地下水运动方程为

$$K_1 M \frac{\partial^2 H}{\partial x^2} + K_2 M \frac{\partial^2 H}{\partial y^2} = S \frac{\partial H}{\partial t}$$

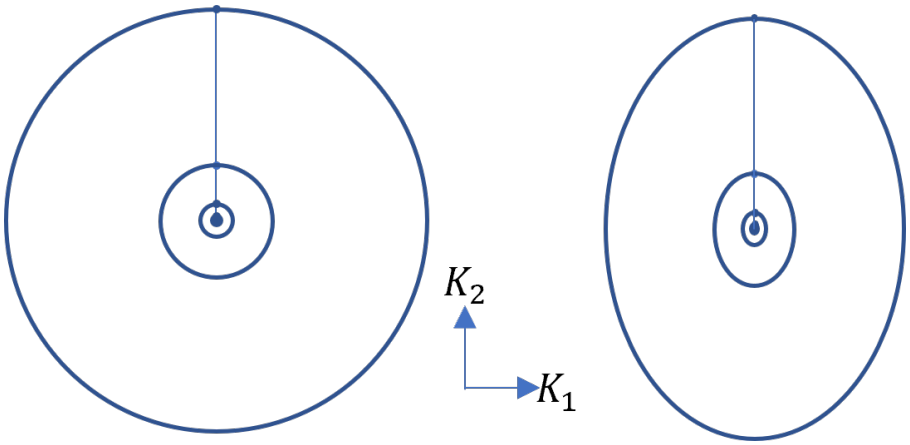
(2) 作坐标变换 $x' = \sqrt{2}x$，$y' = y$，有

$$2K_1 M \frac{\partial^2 H}{\partial x'^2} + K_2 M \frac{\partial^2 H}{\partial y'^2} = S \frac{\partial H}{\partial t}$$

由条件知 $K_2 = 2K_1$，方程变为

$$K_2 M \left(\frac{\partial^2 H}{\partial x'^2} + \frac{\partial^2 H}{\partial y'^2} \right) = S \frac{\partial H}{\partial t}$$

此为均质各项同性含水层承压水运动方程，等水头线为同心圆周。原方程等水头线为 x 方向压缩 $\sqrt{2}$ 后所得的椭圆。



3 .

$$
\begin{cases}
T\left(\frac{\partial^2 H}{\partial x^2}+\frac{\partial^2 H}{\partial y^2}\right)+W=S\frac{\partial H}{\partial t} & (x,y)\in\Omega, \quad t\geq 0 \\
H(x,y,0)=h_0 & (x,y)\in\Omega \\
H(x,y,t)\big|_{\Gamma_{abc}}=H_0 & t>0 \\
T\frac{\partial H}{\partial n}\Big|_{\Gamma_{adc}}=q & t>0 \\
T\frac{\partial H}{\partial r}\Big|_{r_1=r_{w1}}=\frac{Q_1}{2\pi r_{w1}} & t>0 \\
T\frac{\partial H}{\partial r}\Big|_{r_2=r_{w2}}=\frac{Q_2}{2\pi r_{w2}} & t>0
\end{cases}
$$